

Rishabh Tripathi

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github.com/rishabh-tripathi21 | leetcode.com/rishabh_tripathi21 | rishabh21-portfolio.vercel.app

Career Objective

AI/ML enthusiast and Computer Science student with experience in backend development, computer vision, and machine learning. Seeking opportunities in AI/ML Engineering and Backend Development to build scalable, intelligent systems.

Education

Pranveer Singh Institute of Technology, Kanpur Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) Percentage: 74.76% (upto 5th semester)	2023 – Present
R.V.S. International School ISC - 82.2%	2022
R.V.S. International School ICSE - 81.66%	2020

Skills

Programming Languages: Java, Python
AI/ML: Machine Learning, Neural Network, RAG, Agentic AI, NLP, Feature Engineering
Backend: REST APIs, JWT, Authentication, Distributed Systems
Databases: SQL, PostgreSQL
Tools & Platforms: Git, GitHub, VS Code, IntelliJ, Cursor, Postman
Relevant Coursework: Data Structures & Algorithms, DBMS, Computer Network

Projects

CyberTwin <i>Java, Spring Boot, PostgreSQL, Docker, JWT, AI Agents</i> - Designing a multi-agent AI system to simulate cyber attack-defense scenarios with automated threat detection and adaptive response workflows - Building scalable backend services using Spring Boot and PostgreSQL to ingest, process, and query large-scale system event logs in real time - Implementing JWT-based authentication with role-based access control and containerizing services via Docker for reproducible, secure deployments - Architecting a RAG-based threat intelligence layer to enable contextual, real-time decision-making across distributed agent workflows	GitHub Mar 2026 – Present
ANPR – Automatic Number Plate Recognition <i>Python, CNN, BiLSTM, OpenCV, Spring Boot</i> - Built an end-to-end ANPR pipeline combining CNN-based plate detection with BiLSTM character recognition, achieving robust performance across varied lighting and angle conditions - Engineered OpenCV preprocessing — including adaptive thresholding, contour detection, and perspective correction — to normalize plate images before model inference - Deployed the recognition pipeline as a Spring Boot microservice, exposing REST endpoints for real-time inference and enabling seamless integration with traffic or parking management systems	GitHub Oct 2025 – Dec 2025
HandWave – Gesture Controlled System <i>Python, OpenCV, MediaPipe</i> - Developed a real-time hand gesture recognition system achieving smooth touchless control of system functions including volume, brightness, and cursor movement - Engineered a 21-landmark hand tracking pipeline using MediaPipe and OpenCV, mapping gesture patterns to system-level actions with sub-100ms response latency - Optimized frame processing pipeline to sustain 30+ FPS on standard hardware, ensuring responsive and accurate gesture-based interaction	GitHub Aug 2024 – Nov 2024

Certifications

- Spring Framework for Java Development – Coursera, IBM
- Database Management System – NPTEL, IIT Kharagpur
- Machine Learning Specialization – Coursera, Stanford University

Achievements

- Secured 9th rank in college-level hackathon
- Solved 500+ problems across competitive programming platforms including LeetCode and CodeChef